

Supplementary material of the paper
A.M. Karadeniz, N.F. Koçak, C. Hajdu and
L.T. Kóczy

Ahmet Mehmet KARADENİZ^{1,*}, Nuri Furkan KOÇAK², Csaba HAJDU³, and
László T. KÓCZY⁴

¹Doctoral School of Multidisciplinary Engineering Sciences, Széchenyi István University, Győr, Hungary

²Department of Electronics and Automation, Kırıkkale University, Kırıkkale, Türkiye

^{3,4}Department of Informatics, Széchenyi István University, Győr, Hungary

Email: karadeniz.ahmet.mehmet@sze.hu*, nfkocak@kku.edu.tr,
hajdu.csaba@ga.sze.hu, koczy@sze.hu

* Corresponding author

This document provides the supplementary material for the article entitled "*Hamacher Intersection and Optimisation of Aggregation Operators in Fuzzy Signatures for Path Planning*" submitted to the *Romanian Journal of Information Science and Technology*. This file contains detailed experimental setup parameters, extended baseline evaluations, and diagnostic sensitivity plots that support the core metaheuristic optimization framework discussed in the primary manuscript. The digital object identifier (DOI) and article link will be included here upon publication.

Contents of this File

- **Simulation Setup Hyperparameters** — Contains the comprehensive environment configuration constants, optimization weights for the scalar fitness function, and the structural parameters used to initialize the genetic algorithm search population.
- **Canonical Operator Visual Fields** — Displays the reconstructed two-dimensional fuzzy membership fields alongside the corresponding planned trajectories generated by the A^* search algorithm for all eleven uniform operator configurations.
- **Reference Scenario Sensitivity Analysis** — Provides the continuous sensitivity curves for the parametric Hamacher intersection across its discrete parameter grid Γ on the five-obstacle reference scenario for the metric evaluation.

1. Simulation Setup Hyperparameters

Table 3. Configuration constants of the environment, the fitness function and the genetic algorithm.

Group	Parameter	Value
Environment	Root workspace half-width E	12
	Obstacle placement half-width E_o	9
	Grid resolution δ	0.5
	Planning threshold τ	0.5
	Obstacle side lengths $[s_{\min}, s_{\max}]$	$[2, 5]$
	Minimum centre separation	1.0
	Start / goal	$(0, 0) / (10, -10)$
Fitness	Length weight w_L	1.0
	Clearance weight w_c	8.0
	Desired clearance d_{safe}	4.0
	Infeasibility penalty P	10^3
Genetic algorithm	Population size	30
	Generations	25
	Tournament size	3
	Crossover probability	0.8
	Per-gene mutation rate	0.25
	Elitism	2
Hamacher sweep	Parameter grid Γ	$\{0, 0.25, 0.5, 1, 2, 5, 10\}$

2. Canonical Operator Visual Fields

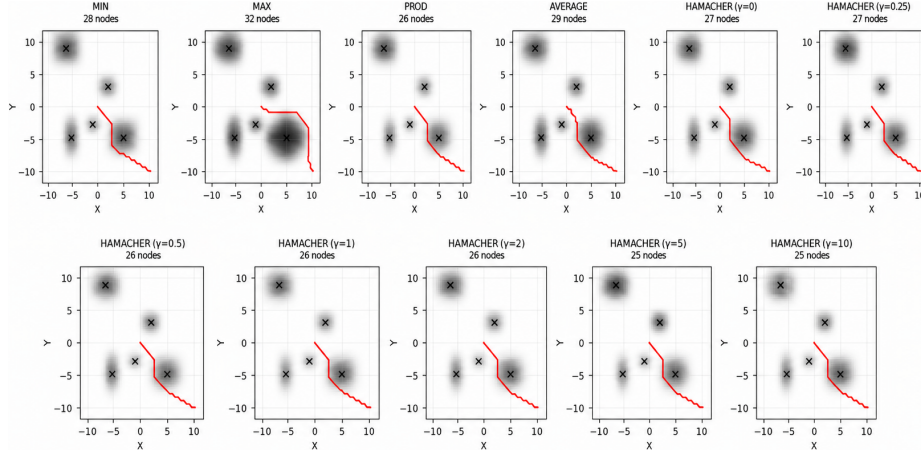


Fig. 4. Membership fields and planned paths of the eleven operator configurations on the five-obstacle reference scenario. Crosses mark obstacle centres; the red polyline is the path planned by A^* .

3. Reference Scenario Sensitivity Analysis

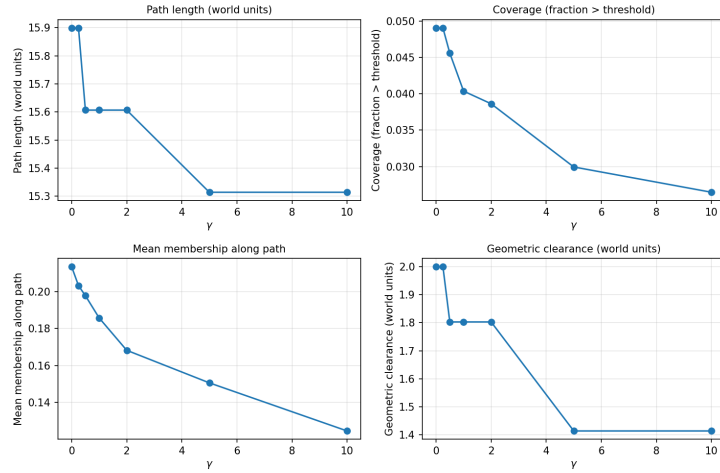


Fig. 5. Sensitivity of path length, coverage, mean membership along the path and geometric clearance to the Hamacher parameter γ on the reference scenario.